Software Engineering
Mini Project

Project Proposal

Project Management System

Group Number: 11

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1. Project Title and Group Number

Project Title: Project Management System

Group Number: 11

2. Team Profile

S.No.	Name	SRN
1	Shivansh Sharma	PES2UG24CS474
2	Shibi R Shetty	PES2UG24CS467
3	Sheetal S Bevinakatti	PES2UG24CS466
4	Shivanagouda I Patil	PES2UG24CS471

3. Proposed Project Description

The Project Management System (PMS) is a web-based application designed to help teams plan, track, and manage their projects efficiently. It combines project workflow management, task/issue tracking, Kanban-style visualization, effort estimation, and sprint/milestone planning into one platform, allowing team members to create tasks, move them through workflow stages, estimate effort, and collaborate through comments and notifications, while project managers get a clear, consolidated view of project progress.

The system addresses problems associated with using disconnected tools (spreadsheets, chat, and email) for project tracking, including lost visibility into task status, inconsistent effort estimates, missed deadlines due to lack of timely notifications, and the absence of a single dashboard summarizing project health. The system is proposed as a new, self-contained academic software project.

Version 1.0 scope: The system supports single-workspace project management — project creation, task/issue tracking, a configurable Kanban board, custom workflows, effort estimation, sprint management, dashboards, notifications, comments, search/filter, reporting, and an activity log. It does not include multi-organization administration, native mobile applications, or deep third-party integrations (e.g., live Slack/GitHub webhooks) beyond a basic REST API; these are considered for future iterations.

Typical Users

- Admin: Manages users, roles, and system-wide settings; has full access to all projects and audit logs.
- Project Manager: Creates and configures projects, defines workflows, manages sprints, and reviews dashboards/reports.
- Team Member: Creates, updates, and moves tasks assigned to them, comments, and collaborates within a project.

Functional Features

- Secure login, logout, password reset, and role-based access control (Admin, Project Manager, Team Member).
- Create, edit, archive, and delete projects, and add or remove team members.
- Create, edit, delete, and assign tasks with title, description, priority, due date, and labels.

- Visualize tasks on a Kanban board (Backlog, To Do, In Progress, Review, Done) with drag-and-drop status updates.
- Define custom workflow states and valid transitions between them for each project.
- Assign story points or time estimates to tasks and track estimated versus actual effort.
- Create sprints/milestones, add tasks to a sprint backlog, and close sprints.
- View a dashboard summarizing project progress, burndown/velocity charts, and task distribution.
- Receive notifications on task assignment, status changes, comments, and approaching deadlines.
- Comment on tasks and attach files or documents for collaboration.
- Search and filter tasks or projects by assignee, priority, status, label, or sprint.
- Generate downloadable reports such as progress summaries, velocity, and workload.
- Maintain an audit trail (activity log) of changes made to a task or project.

4. Plan of Work and Product Ownership

4.1 Plan of Work (Short Term)

Week	Planned Activity
1–2	Finalize the SRS: requirements gathering, functional/non-functional requirement documentation, ER diagram and database schema design.
3	Design UML use-case and class diagrams; set up project repository, technology stack and development environment.
4–5	Implement core modules: user authentication with role-based access control, and project management (create/edit/archive/delete projects, team membership).
6	Implement task/issue management, the Kanban board with drag-and-drop, and configurable workflow states/transitions.
7	Implement effort estimation and sprint/milestone management (create sprint, backlog, close sprint).
8	Implement the dashboard (progress, burndown/velocity, task distribution), notifications, comments/collaboration, and search/filter.
9	Implement reporting and the activity log; integration testing across modules and UI refinement.
10	Final testing, bug fixes, documentation and demo preparation.

4.2 Product Ownership

The team consists of four members. Functional-feature ownership is assigned according to the major functional features of the proposed system. UI/graphic design, database interaction, unit testing and other common project activities are shared responsibilities across the team and are not counted as functional features.

Team Member	Functional Feature(s) Owned	Qualitative Goal
Shivansh Sharma (Team Lead)	Login, roles and access control; Project Management (create/edit/archive/delete, team membership); Activity Log.	Ensure protected operations and correct role-based access across every project, and maintain a complete, tamper-evident audit trail of project and task changes.

Team Member	Functional Feature(s) Owned	Qualitative Goal
Shibi R Shetty	Task/Issue Management; Kanban Board; Workflow Management.	Ensure smooth drag-and-drop Kanban updates and that task status changes strictly follow each project's defined workflow transitions.
Sheetal S Bevinakatti	Estimation; Sprint/Milestone Management; Notifications; Comments & Collaboration.	Keep estimated-vs-actual effort tracking accurate across sprints, and ensure notifications and comments are delivered promptly so collaboration stays in sync.
Shivanagouda I Patil	Visibility & Dashboard; Search & Filter; Reporting.	Ensure the dashboard, search/filter, and generated reports accurately reflect current project data through a clear, easy-to-read UI, with search/filter results returned within 2 seconds.

Scope Clarification

The project focuses on single-workspace project, task, and sprint management with Kanban visualization, estimation, dashboards, notifications, and reporting. Multi-organization/workspace administration, native mobile apps, and deep real-time third-party integrations are not part of Version 1.0 and are noted as future scope.